Grade 5

What does GS73 suggest about life at Madjedbebe in ancient Australia?

■ Origin, features & purpose | ▲ Historical context | ● Historical interpretations | ◆ Accuracy, usefulness & reliability

WORKED EXAMPLE

GS73 is a **stone source** found at Madjedbebe. I can see **deep grooves** and **shiny patches**. It comes from **ancient Australia** and may have been used for grinding. Researchers **disagree about exactly how old its soil layer is**. The stone is **useful for showing tool use**, but it **cannot tell us the user's name**.

WHAT EACH MARKED PHRASE PROVES

- ■ Names the source and two visible features: 'stone source', 'deep grooves' and 'shiny patches'. Origin and purpose are not yet developed.
- ▲ Places GS73 broadly in 'ancient Australia', but gives little detail about life or practice.
- ● Notices disagreement about the date without describing the competing arguments.
- ◆ Gives one use and one limit: tool use is visible, but the user cannot be named.

LEARNING CONTINUUM AT GRADE 5

■ ORIGIN, FEATURES & PURPOSE	I can name a historical source and a feature I notice.
▲ HISTORICAL CONTEXT	I can say where or when a source comes from.
● HISTORICAL INTERPRETATIONS	I can notice that people may explain the past differently.
◆ ACCURACY, USEFULNESS & RELIABILITY	I can say one thing a source can and cannot tell us.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), "65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia," *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.

Year 6

What does GS73 suggest about life at Madjedbebe in ancient Australia?

■ Origin, features & purpose | ▲ Historical context | ● Historical interpretations | ◆ Accuracy, usefulness & reliability

WORKED EXAMPLE

GS73 is a stone tool: a broken piece of a grinding stone from Madjedbebe, on Mirarr Country. Photographs show deep grooves and shiny, polished patches. People were living there many thousands of years ago, and they used stones like this one to grind their food. Some archaeologists believe the layer is about 65,000 years old. Others believe the soil has moved, so the date is not certain. The stone is useful evidence that people were grinding materials, but it cannot tell us which plants they were grinding.

WHAT EACH MARKED PHRASE PROVES

- ■ ‘Archaeological artefact’, ‘sandstone grinding-stone fragment’ and ‘grooves and polish’ identify type and features; purpose is not yet explained.
- ▲ ‘A very early period of Aboriginal occupation’ identifies the context and connects it to repeated tool use.
- ● Recognises two positions on the date, but does not yet describe the evidence behind them.
- ◆ Recognises usefulness for grinding and a clear limit: the material processed cannot yet be identified.

LEARNING CONTINUUM AT YEAR 6

■ ORIGIN, FEATURES & PURPOSE	I can identify different types of historical sources.
▲ HISTORICAL CONTEXT	I can identify the historical context of a source.
● HISTORICAL INTERPRETATIONS	I can recognise that historians may interpret historical events and developments differently.
◆ ACCURACY, USEFULNESS & RELIABILITY	I can recognise that some historical sources are more useful than others.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), “65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia,” *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.

Year 7 EXPECTED

What does GS73 suggest about life at Madjedbebe in ancient Australia?

■ Origin, features & purpose | ▲ Historical context | ● Historical interpretations | ◆ Accuracy, usefulness & reliability

WORKED EXAMPLE

The ancient object is GS73, a broken piece of a sandstone grinding stone excavated at Madjedbebe, on Mirarr Country. The modern source is a set of photographs published by Hayes and her team in 2022. Their purpose was to record the wear and argue that people ground food there for a very long time. In ancient Australia, grinding allowed people to eat hard seeds and tough plants. It also shows they understood the resources around them. Hayes and her team interpret this as plant processing, because the stone carries deep grooves, microscopic polish and 41 starch grains. Other researchers question how old the layer really is. GS73 is very useful for studying how people prepared food, because its shape, its wear and its plant traces all agree, but it cannot tell us which plant was ground.

WHAT EACH MARKED PHRASE PROVES

- ■ Separates the ancient object from the modern 2022 record, then identifies origin, features and why the figure was created.
- ▲ Outlines how grinding supported plant preparation and knowledge of local resources.
- ● Describes the plant-processing interpretation and the separate challenge to the earliest date.
- ◆ Concludes that converging methods make plant processing useful evidence, then bounds the claim with two limits.

LEARNING CONTINUUM AT YEAR 7

■ ORIGIN, FEATURES & PURPOSE	I can identify the origin, content features and purpose of historical sources.
▲ HISTORICAL CONTEXT	I can outline the historical context of a source.
● HISTORICAL INTERPRETATIONS	I can describe different historical interpretations of the past.
◆ ACCURACY, USEFULNESS & RELIABILITY	I can draw conclusions about the usefulness of sources.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), “65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia,” *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.

Year 8

What does GS73 suggest about life at Madjedbebe in ancient Australia?

■ Origin, features & purpose | ▲ Historical context | ● Historical interpretations | ◆ Accuracy, usefulness & reliability

WORKED EXAMPLE

GS73 is a broken millstone recovered from Phase 2, one of the oldest occupation layers at Madjedbebe. Hayes and her team photographed it and examined it under a microscope. They published the images so that other researchers could inspect the same evidence and judge their interpretation. Madjedbebe is a rock shelter on Mirarr Country that was occupied for many thousands of years. The grinding marks show that people were already preparing plant foods early in Aboriginal history. Hayes explains the wear and the plant residues as signs of seed grinding. Clarkson argues that the artefacts remained where they were dropped, which supports the early date. Williams points to termite activity and moving sediment, which would make that date less reliable. The shape, the microscopic wear, the starch and the plant chemicals all point in the same direction, so the grinding interpretation is well supported. Even so, I would ask whether the residues could be contaminated, and whether the layer could have shifted.

WHAT EACH MARKED PHRASE PROVES

- ■ Describes the object and explains how the modern figure lets readers inspect its form, wear and residues.
- ▲ Describes Madjedbebe, Mirarr Country and grinding within deep Aboriginal history.
- ● Explains the debate by connecting each interpretation to a different evidence base.
- ◆ Compares independent evidence for function, then asks targeted reliability questions about contamination and layer movement.

LEARNING CONTINUUM AT YEAR 8

■ ORIGIN, FEATURES & PURPOSE	I can describe the origin, content features and purpose of historical sources.
▲ HISTORICAL CONTEXT	I can describe the historical context of sources.
● HISTORICAL INTERPRETATIONS	I can explain different historical interpretations and contested debates about the past.
◆ ACCURACY, USEFULNESS & RELIABILITY	I can compare and contrast historical sources and ask questions about their accuracy, usefulness and reliability.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), “65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia,” *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.

Year 9

What does GS73 suggest about life at Madjedbebe in ancient Australia?

■ Origin, features & purpose | ▲ Historical context | ● Historical interpretations | ◆ Accuracy, usefulness & reliability

WORKED EXAMPLE

GS73 is material evidence, excavated from a recorded layer called Phase 2. Hayes and her team published photographs, microscope images and laboratory results together. They did this because one form of evidence alone would not convince a scientific audience. The study belongs to wider research into the deep history of Aboriginal Australia, and it was carried out with the permission of Mirarr custodians. Hayes analyses the grooves, the polish and the plant residues as evidence of seed processing. Clarkson analyses the condition of the site and the dating to defend the early chronology. Williams uses signs of termite disturbance and sediment movement to challenge it. Because the wear and the residues agree independently, grinding is the interpretation I would trust most. The exact plant and the precise age are weaker, because both depend on preservation and on how stable the deposit is.

WHAT EACH MARKED PHRASE PROVES

- ■ Explains why the peer-reviewed figure combines several records, and which audience and argument it addresses.
- ▲ Explains the deep-history research context and the role of Mirarr custodial permission.
- ● Analyses each interpretation through its evidence: wear and residue, site integrity and dating, or termite and sediment movement.
- ◆ Identifies grinding as a stronger claim than exact plant or exact age, and explains why.

LEARNING CONTINUUM AT YEAR 9

■ ORIGIN, FEATURES & PURPOSE	I can explain the origin, content features and purpose of historical sources.
▲ HISTORICAL CONTEXT	I can explain historical context, including important events that influenced a source’s creation.
● HISTORICAL INTERPRETATIONS	I can analyse historical interpretations and contested debates and identify the evidence used to support them.
◆ ACCURACY, USEFULNESS & RELIABILITY	I can compare and contrast historical sources and identify their accuracy, usefulness and reliability.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), “65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia,” *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.

Year 10

What does GS73 suggest about life at Madjedbebe in ancient Australia?

■ Origin, features & purpose	▲ Historical context	● Historical interpretations	◆ Accuracy, usefulness & reliability

WORKED EXAMPLE

GS73 is an ancient artefact, but we meet it only through a modern scientific publication. Its explicit evidence is the recorded wear and the plant residues preserved on the stone. Its implicit argument is that grinding technology in Australia is far older than earlier accounts allowed, and that argument is aimed at other archaeologists. The investigation reflects modern laboratory testing carried out with Mirarr custodians. It also reveals what those researchers value: testing claims, seeking permission, and stating honestly how certain they are. Hayes and Clarkson accept the early chronology because several independent records converge on it. Williams gives more weight to termite disturbance and sediment movement. Their interpretations differ because they evaluate the formation of the site differently. Overall, GS73 is strong evidence for grinding, but only qualified evidence for a precise age. Its main strength is that separate tests agree; its limits are the unidentified plant, the modern plastic contamination, and the ongoing argument about the layer.

WHAT EACH MARKED PHRASE PROVES

- ■ Analyses explicit evidence, the publication’s implicit argument, its purpose and its scientific audience.
- ▲ Uses context to explain the creators’ values: empirical testing, custodial permission and careful qualification.
- ● Exposes why interpretations differ — researchers give different weight to converging records and site-formation evidence.
- ◆ Weighs a strength, three limitations, and two claims that carry different degrees of certainty.

LEARNING CONTINUUM AT YEAR 10

■ ORIGIN, FEATURES & PURPOSE	I can analyse the origin, explicit and implicit meaning, and purpose of historical sources.
▲ HISTORICAL CONTEXT	I can explain historical context to identify the motivations, values and attitudes of a source’s creator.
● HISTORICAL INTERPRETATIONS	I can evaluate historical interpretations and contested debates, including why or how interpretations differ.
◆ ACCURACY, USEFULNESS & RELIABILITY	I can compare and contrast sources and analyse their accuracy, usefulness and reliability.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), “65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia,” *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.